AWS Lambda — Important Interview Questions & Answers

1.What is AWS Lambda?
AWS Lambda is a serverless compute service that runs code without requiring us to manage servers. We can run code in response to events and AWS automatically manages the underlying infrastructure.
Simple ga gurthupettuko:
Lambda ante servers manage cheyyakunda code run cheyyadaniki use chese serverless service.
Eng: AWS Lambda is a serverless service that runs code without managing servers.

2.What is a Lambda Function?
A Lambda Function is the code and configuration that AWS Lambda runs when it is invoked. It contains the application logic required to perform a specific task.
Simple ga gurthupettuko:
Lambda Function ante specific task perform cheyyadaniki Lambda lo run ayye code.
Eng: A Lambda Function contains code that runs when the function is invoked.

3.What are Lambda Versions?
Lambda Versions are immutable snapshots of a Lambda function's code and configuration. Each published version gets a unique version number.
Simple ga gurthupettuko:
Version ante Lambda function code and configuration yokka fixed snapshot.
Eng: A Lambda Version is an immutable snapshot of a function's code and configuration.

4.What is a Lambda Alias?
A Lambda Alias is a name that points to a specific Lambda function version. It helps us manage different environments such as development, testing, and production.
Simple ga gurthupettuko:
Alias ante specific Lambda version ni easy name tho refer cheyyadaniki use chestham.
Eng: A Lambda Alias points to a specific version of a Lambda function.

5.What are Lambda Triggers?
Lambda Triggers are events or AWS services that invoke a Lambda function. Examples include S3 events, API Gateway requests, EventBridge events, and SQS messages.
Simple ga gurthupettuko:
Trigger ante Lambda function ni automatically invoke chese event or service.
Eng: A Lambda Trigger is an event or service that invokes a Lambda function.

6.What are Lambda Permissions?
Lambda Permissions control who or what can invoke a Lambda function. Permissions can be configured using resource-based policies and IAM policies depending on the access scenario.
Simple ga gurthupettuko:
Lambda Permissions ante evaru Lambda function ni invoke cheyyalo control cheyyadam.
Eng: Lambda Permissions control who or what can invoke a Lambda function.

7.What are Lambda Environment Variables?
Environment Variables are key-value settings available to a Lambda function during execution. They are commonly used to store configuration values that can change between environments.
Simple ga gurthupettuko:
Environment Variables ante Lambda code lo direct ga hardcode cheyyakunda configuration values store cheyyadaniki use chestham.
Eng: Environment Variables store configuration values used by a Lambda function.

8.What are Lambda Layers?
Lambda Layers are packages containing libraries, dependencies, or other files that can be shared across multiple Lambda functions. They help reduce duplicate code and simplify dependency management.
Simple ga gurthupettuko:
Layer ante common libraries and dependencies ni multiple Lambda functions ki share cheyyadaniki use chestham.
Eng: Lambda Layers allow libraries and dependencies to be shared across functions.

9.What is Lambda Concurrency?
Concurrency is the number of Lambda function instances that can run at the same time. AWS Lambda automatically scales function executions based on incoming requests, subject to concurrency limits.
Simple ga gurthupettuko:
Concurrency ante same time lo enni Lambda executions run avuthunnayo.
Eng: Concurrency is the number of Lambda executions running at the same time.

10.What is Reserved Concurrency?
Reserved Concurrency sets a specific concurrency limit for a Lambda function and reserves that amount of concurrency for the function.
Simple ga gurthupettuko:

Reserved Concurrency ante Lambda function ki concurrency ni reserve and limit cheyyadam.
Eng: Reserved Concurrency reserves and limits concurrency for a Lambda function.

11.What is Provisioned Concurrency?
Provisioned Concurrency keeps a specified number of execution environments initialized and ready to respond to requests. It helps reduce cold start latency.
Simple ga gurthupettuko:
Provisioned Concurrency ante Lambda execution environments ni mundugane ready ga maintain cheyyadam.
Eng: Provisioned Concurrency keeps Lambda execution environments ready to reduce cold start latency.

12.What is a Lambda Cold Start?
A cold start happens when Lambda needs to initialize a new execution environment before running a function. This initialization can add some extra latency to the first request.
Simple ga gurthupettuko:
Cold Start ante new Lambda environment initialize avvadam valla first request ki extra time padatam.
Eng: A Lambda Cold Start occurs when a new execution environment must be initialized.

13.What is Lambda Monitoring?
Lambda Monitoring helps us view function performance, errors, invocations, duration, and other operational information. Amazon CloudWatch is commonly used to monitor Lambda functions and their logs.
Simple ga gurthupettuko:
Lambda Monitoring ante function performance, errors, logs and executions ni check cheyyadam.
Eng: Lambda Monitoring helps track Lambda performance, errors, invocations, and logs.

14.What is a Lambda Invocation?
Invocation is the process of running a Lambda function. A function can be invoked synchronously, asynchronously, or through event source mappings depending on the integration.
Simple ga gurthupettuko:
Invocation ante Lambda function ni execute or run cheyyadam.
Eng: Invocation is the process of running a Lambda function.

15.What is the difference between Synchronous and Asynchronous Invocation?
In synchronous invocation, the caller waits for the Lambda function response. In asynchronous invocation, Lambda accepts the event and processes it without requiring the caller to wait for the function result.
Simple ga gurthupettuko:
Synchronous lo caller response kosam wait chestadu, Asynchronous lo wait cheyyakunda event process avutundi.
Eng: Synchronous invocation waits for a response, while asynchronous invocation does not require the caller to wait.

16.What is the Lambda Execution Role?
A Lambda Execution Role is an IAM Role that gives the Lambda function permission to access AWS resources during execution.
Simple ga gurthupettuko:
Execution Role ante Lambda function AWS resources ni access cheyyadaniki use chese IAM Role.
Eng: A Lambda Execution Role provides permissions for a Lambda function to access AWS resources.

17.How does Lambda access S3?
Lambda can access S3 using an IAM Execution Role with the required S3 permissions. The Lambda function uses these permissions to perform actions such as reading or writing objects.
Simple ga gurthupettuko:
Lambda ki S3 access kavali ante Execution Role lo required S3 permissions ivvali.
Eng: Lambda accesses S3 using an execution role with the required permissions.

18.Can Lambda access a database?
Yes. Lambda can connect to supported databases when the required networking, authentication, and permissions are configured. For databases inside a VPC, the Lambda function can be configured to access the required VPC resources.
Simple ga gurthupettuko:
Lambda database ki connect avvachu, required network and authentication settings correct ga unte.
Eng: Lambda can connect to databases when the required network and access configuration is available.

19.What happens when a Lambda function is invoked many times at the same time?
Lambda can create multiple execution environments to handle concurrent requests. The number of executions is controlled by available and configured concurrency limits.
Simple ga gurthupettuko:
Same time lo many requests vaste Lambda multiple execution environments create chesi handle chestundi.
Eng: Lambda can create multiple execution environments to handle concurrent requests.

20.How would you troubleshoot a Lambda function failure?
I would check CloudWatch logs, Lambda metrics, function configuration, IAM permissions, environment variables, dependencies, triggers, and the error message. If the function accesses a VPC resource, I would also check the networking configuration.
Simple ga gurthupettuko:
Lambda fail ayite Logs → Metrics → IAM → Environment Variables → Dependencies → Trigger → Network check chestham.
Eng: For a Lambda failure, I would check logs, metrics, permissions, configuration, triggers, and networking.

21.What is the difference between Lambda Versions and Aliases?
A Lambda Version is a fixed snapshot of the function, while an Alias is a named pointer to a specific version. An alias can be updated to point to another version without changing the alias name.
Simple ga gurthupettuko:
Version fixed snapshot, Alias aa version ni point chese easy name.
Eng: A Version is a fixed snapshot, while an Alias points to a specific version.

22.Why do we use Lambda Layers?
We use Lambda Layers to package and share common libraries, dependencies, and files across multiple functions. This helps reduce duplicate packages and makes function deployment easier.
Simple ga gurthupettuko:
Multiple functions lo same dependencies unte Layer use chesi common ga share chestham.
Eng: Lambda Layers help share common dependencies across multiple functions.

23.Why do we use Environment Variables?
We use Environment Variables to keep configuration values separate from the function code. This makes it easier to use different settings for development, testing, and production environments.
Simple ga gurthupettuko:
Different environments ki different configuration values use cheyyadaniki Environment Variables use chestham.
Eng: Environment Variables separate configuration from Lambda function code.

24.Why do we use Lambda?
We use Lambda to run code without managing servers. It is useful for event-driven applications, automation, APIs, background processing, and many other workloads.
Simple ga gurthupettuko:
Servers manage cheyyakunda event-based code run cheyyadaniki Lambda use chestham.
Eng: Lambda runs code without requiring us to manage servers.

25.What is the most important thing to remember about AWS Lambda?
AWS Lambda is a serverless compute service that runs code in response to events without requiring server management. The important concepts are Functions, Versions, Aliases, Triggers, Permissions, Environment Variables, Layers, Concurrency, and Monitoring.
Simple ga gurthupettuko:
Lambda ante servers manage cheyyakunda code run chese serverless service.
Eng: Lambda is a serverless compute service used to run code without managing servers.